

Solubility

Solubility: The maximum amount of a particular solute which will dissolve in a particular solvent at a specific temperature.

Solubility is often expressed in the maximum mass of solute in grams that will dissolve in a solvent (measured in grams or kg) at a given temperature/pressure (typically SATP).

Some examples of the solubility of common substances at 25°C:

Substance	Solubility (g/100g solution)
iron(III) chloride	73.1
lead iodide	0.08
lithium sulfate	27.2
silver nitrate	71.5
sodium sulfate	27.7

There are 3 possibilities for the condition of a solution:

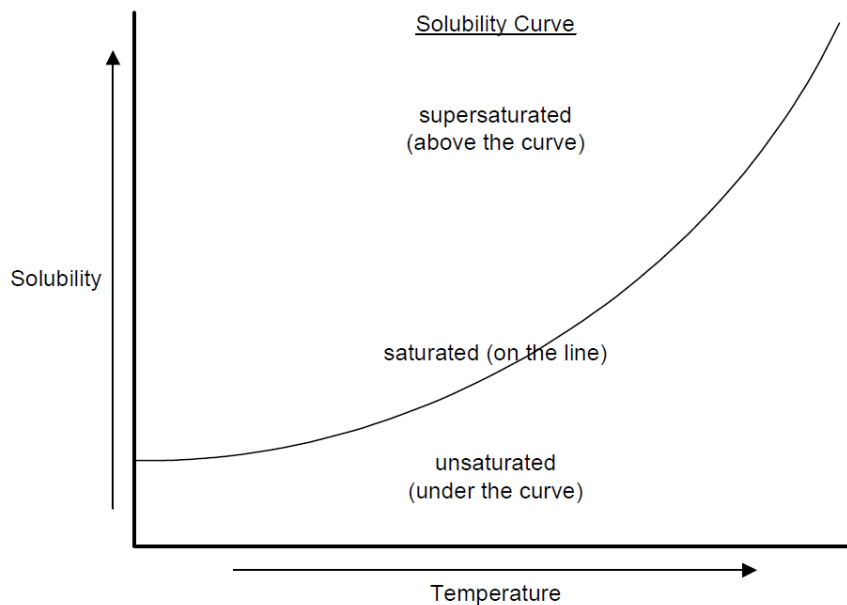
Unsaturated: there is less than the maximum amount of solute dissolved.

Saturated: the maximum amount of solute is present in solution

Supersaturated: the amount of solute exceeds at a specific temperature exceeds its solubility.

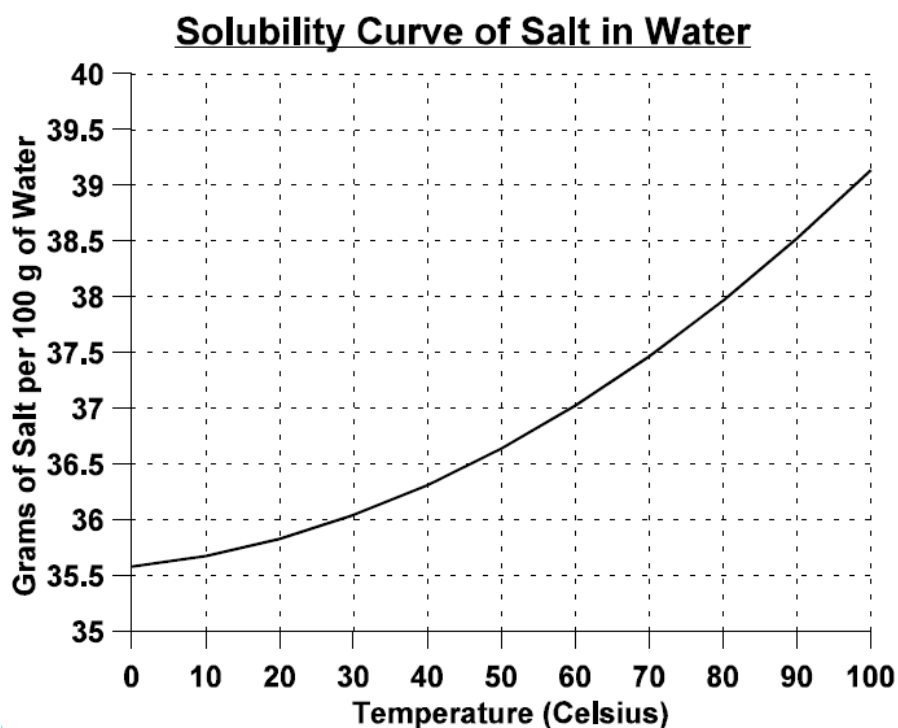
Supersaturated solutions must be prepared by first fully dissolving the solute at a higher temperature, and then cooling the solution slowly. This is not always possible: most times the excess solute will precipitate out of the solution as it cools.

A solubility curve shows how the solubility changes with temperature. At a given temperature, the solubility values below the curve indicate unsaturated solutions. The point on the curve indicates a saturated solution. The points above the curve indicate supersaturated solutions.



Questions: Use the solubility curve below to answer the following questions

- State fully the solubility of salt at these temperatures:
 - 40°C
 - 65°C
 - 90°C
- At what temperature would these solutions be saturated?
 - 36.5 g of salt in 100 g of water
 - 76.5 g of salt in 200 g of water
 - 19.0 g of salt in 50 g of water
- State whether the following solutions are saturated, unsaturated or supersaturated.
 - 37.0 g of salt in 100 g of water at 70°C
 - 37.0 g of salt in 100 g of water at 45°C
 - 111.0 g of salt in 300 g of water at 90°C
- Determine the mass of salt that should precipitate when a saturated solution at 90°C is allowed to cool to 10°C.



Answers:

1.a) 36.5 g/kg b) 37.5 g/kg c) 38.5 g/kg 2.a) 45°C b) 85°C c) 80°C 3.a) unsaturated b) supersaturated c) unsaturated 4. 2.9 g/100g of water